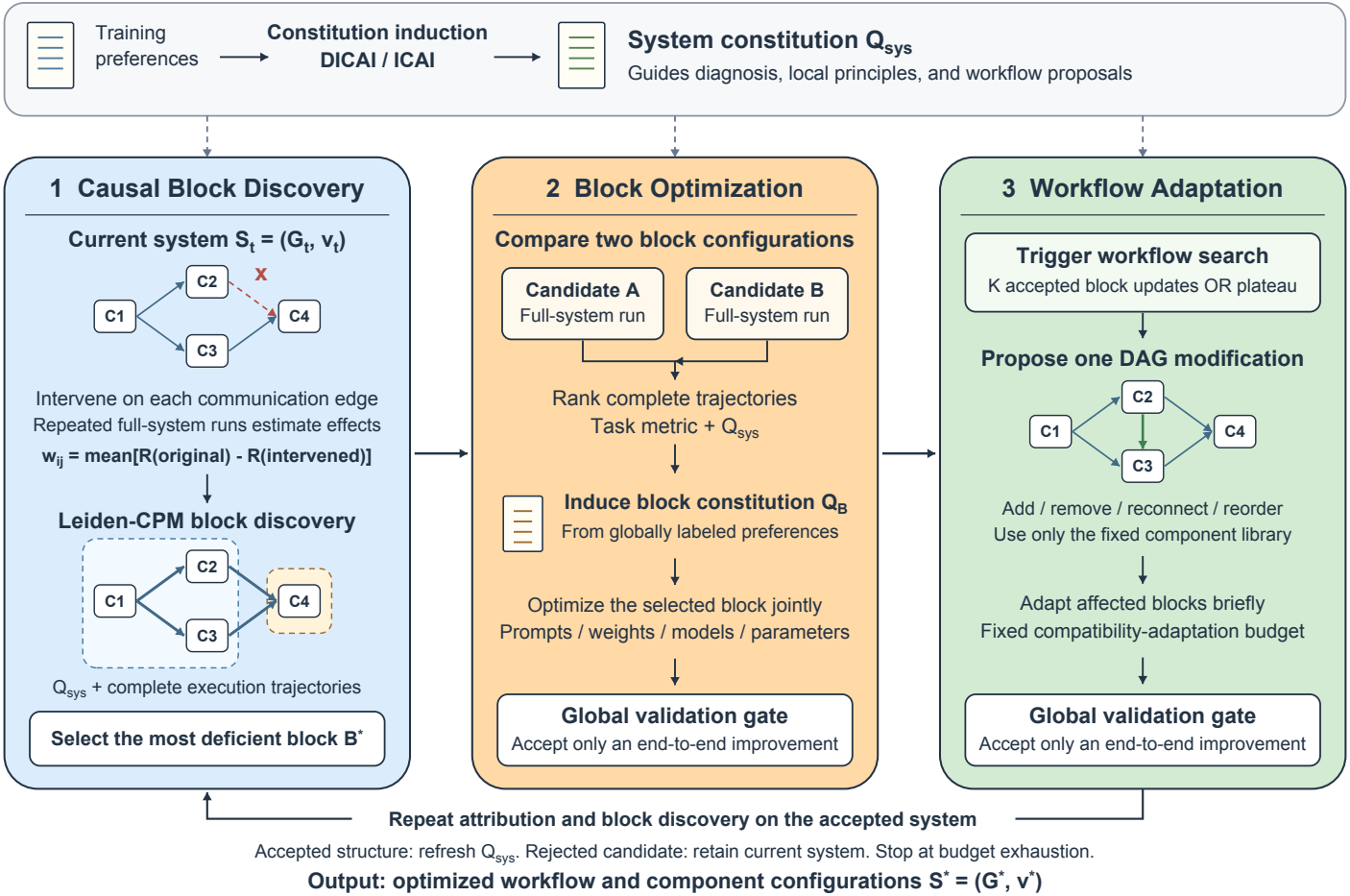

Training preferences
Constitution induction
DICAI / ICAI
System constitution Q_{sys}
Guides diagnosis, local principles, and workflow proposals
1 Causal Block Discovery
Current system $S_t = (G_t, v_t)$
C1
C2
C3
C4
X
Intervene on each communication edge
Repeated full-system runs estimate effects
w_{ij} = mean[R(original) - R(intervened)]
Leiden-CPM block discovery
Q_{sys} + complete execution trajectories
Select the most deficient block B^*
2 Block Optimization
Compare two block configurations
Candidate A
Full-system run
Candidate B
Full-system run
Rank complete trajectories
Task metric + Q_{sys}
Induce block constitution Q_B
From globally labeled preferences
Optimize the selected block jointly
Prompts / weights / models / parameters
Global validation gate
Accept only an end-to-end improvement
3 Workflow Adaptation
Trigger workflow search
K accepted block updates OR plateau
Propose one DAG modification
Add / remove / reconnect / reorder
Use only the fixed component library
Adapt affected blocks briefly
Fixed compatibility-adaptation budget
Global validation gate
Accept only an end-to-end improvement
Repeat attribution and block discovery on the accepted system
Accepted structure: refresh Q_{sys}. Rejected candidate: retain current system. Stop at budget exhaustion.
Output: optimized workflow and component configurations $S^* = (G^*, v^*)$